👩🏾‍🏫 Staff Guide · Chemistry · Unit 1B Electric Forces (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Mr. Brittle's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 1 Lightning: L1.7 Coulomb's Law (+ scientific notation, sig-fig extension), L1.8 Neutral Interactions, L1.9 Static Interaction Review")	**Level 2 piece**
L1.7 Coulomb's Law (PhET sim + calculations with k, scientific notation, sig-fig extension)	Slides "Closer is stronger", "More charge is stronger" + notes part B. Level 1-2: picture comparisons, no formula. Level 3: proportional reasoning
L1.7 Scientific Notation Explanation	Level 2 slide "Big numbers" + notes part C; Level 3 math practice
L1.8 Neutral Interactions (PhET balloon sim, paper scraps)	Slides "Balloon and paper" / "Neutral objects" + notes part C
L1.9 Static Interaction Review	Unit quiz below
Class slides	Deck: 18 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 14 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: 🧲 pull, ↔️ push, 🤏 close = 💪 strong, 🔋 more charge = 💪 strong
- No calculations at Level 1 or 2; the sig-fig extension is skipped

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- L1.7 calculations: Level 1-2 students answer the picture comparisons instead (same points when complete)
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Mr. Brittle".

6. Photo credits (open licenses)

Attractive-electric-force-between-hair-and-balloon.jpg — MikeRun, CC BY-SA 4.0, Wikimedia Commons

Lightning lessons 7-9 notes

Chemistry · Unit 1B Electric Forces · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Attract and repel (L1.7, L1.9)

1. Opposite charges _____.
2. Like charges _____.
3. Circle the pairs that repel:

− and − | + and − | + and +

Word bank: **attract** **repel**

B. Coulomb's law (L1.7)

4. More charge makes the force _____.
5. More distance makes the force _____.
6. Match the change to what happens:

double the charge	force gets stronger
double the distance	force gets 4× weaker
move closer	force gets 2× stronger

Word bank: **stronger** **weaker**

C. Big numbers and neutral objects (L1.7, L1.8)

7. 2×10^3 means 2 with three _____: 2,000.
8. A neutral object near a charged balloon becomes _____.
9. Number the steps (1-4):

_____ The balloon gets extra electrons

_____ The paper jumps to the balloon

_____ Rub the balloon on hair

_____ Paper's electrons shift away

Word bank: **polarized** **zeros**

Level 2 version of the gen ed "L1.7 Coulomb's Law, L1.8 Neutral Interactions, L1.9 Review."

📖 Electric Forces: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Attract

Pull toward each other. Opposite charges (+ and −) attract.



Repel

Push away from each other. Like charges (− and −, or + and +) repel.



Electric force

The push or pull between charged objects. It works without touching.



Distance

How far apart two objects are. Closer = stronger force.



Amount of charge

How many extra or missing electrons. More charge = stronger force.



Polarized

Electrons in a neutral object shift to one side, so one side is a little − and the other a little +.



Coulomb's law

The rule for electric force: more charge makes it stronger; more distance makes it much weaker.



Scientific notation

A short way to write very big or small numbers, like $2 \times 10^{12} = 2{,}000{,}000{,}000{,}000$.

✅ Electric Forces Quiz

Chemistry · Unit 1B · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🧲 Opposite charges…

A. Repel | B. Attract | C. Cancel the atoms

2. ↔️ Two negative balloons will…

A. Attract | B. Become neutral | C. Repel

3. 🔋 Which makes the electric force stronger?

A. More charge | B. More distance | C. Less charge

4. 📏 If the distance gets bigger, the force gets…

A. Stronger | B. The same | C. Weaker

5. ⚠️ Double the distance, and the force becomes…

A. ¼ as strong | B. 2 times stronger | C. The same

6. 🔢 3×10^4 is the same as…

A. 3,000 | B. 30,000 | C. 34

7. 📈 Which number is bigger?

A. 5×10^9 | B. 5×10^{12} | C. 5×10^3

8. 🔀 A balloon pulls on neutral paper because the paper becomes…

A. Polarized | B. Positive forever | C. Magnetic

9. 🙌 Electric forces…

A. Only work when touching | B. Only work on metal | C. Work without touching

10. 🎈 A balloon rubbed for longer gets…

A. More charge and a stronger pull | B. Less charge | C. No charge

Lightning lessons 7-9 notes · ANSWER KEY

Chemistry · Unit 1B Electric Forces · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Attract and repel (L1.7, L1.9)

1. Opposite charges **attract**.
2. Like charges **repel**.
3. Circle the pairs that repel:

− and − ＋ and − **＋ and ＋**

Word bank: **attract** **repel**

B. Coulomb's law (L1.7)

4. More charge makes the force **stronger**.
5. More distance makes the force **weaker**.
6. Match the change to what happens:

(Key: double the charge → force gets 2× stronger · double the distance → force gets 4× weaker · move closer → force gets stronger)

double the charge	force gets stronger
double the distance	force gets 4× weaker
move closer	force gets 2× stronger

Word bank: **stronger** **weaker**

C. Big numbers and neutral objects (L1.7, L1.8)

7. 2×10^3 means 2 with three **zeros**: 2,000.
8. A neutral object near a charged balloon becomes **polarized**.
9. Number the steps (1-4):

2 The balloon gets extra electrons
4 The paper jumps to the balloon
1 Rub the balloon on hair
3 Paper's electrons shift away

Word bank: **polarized** **zeros**

Level 2 version of the gen ed "L1.7 Coulomb's Law, L1.8 Neutral Interactions, L1.9 Review."
🔑 Key written by our team from Mr. Brittle's materials, please check.

✅ Electric Forces Quiz · ANSWER KEY

Chemistry · Unit 1B · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🧲 Opposite charges… L1.9

A. Repel | **B. Attract** | C. Cancel the atoms

2. ↔️ Two negative balloons will… L1.9

A. Attract | B. Become neutral | **C. Repel**

3. 🔋 Which makes the electric force stronger? L1.7

A. More charge | B. More distance | C. Less charge

4. 📏 If the distance gets bigger, the force gets… L1.7

A. Stronger | B. The same | **C. Weaker**

5. ⚠️ Double the distance, and the force becomes… L1.7

A. ¼ as strong | B. 2 times stronger | C. The same

6. 🔢 3×10^4 is the same as… L1.7

A. 3,000 | **B. 30,000** | C. 34

7. 📈 Which number is bigger? L1.7

A. 5×10^9 | **B. 5×10^{12}** | C. 5×10^3

8. 🔀 A balloon pulls on neutral paper because the paper becomes… L1.8

A. Polarized | B. Positive forever | C. Magnetic

9. 🙌 Electric forces… L1.8

A. Only work when touching | B. Only work on metal | **C. Work without touching**

10. 🎈 A balloon rubbed for longer gets… L1.14

A. More charge and a stronger pull | B. Less charge | C. No charge

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "L1.9 Static Interaction Review + L1.14 Practice Test (questions 2-6)". Score: __ / 10.

🔑 *Key written by our team from Mr. Brittle's materials, please check.*